Exp. No: 09 Implementation of Subnetting in Cisco Packet Tracer Simulator.
Date: 8/9/26

Design & configure a Simple Internetwork using a Router.

AIM:

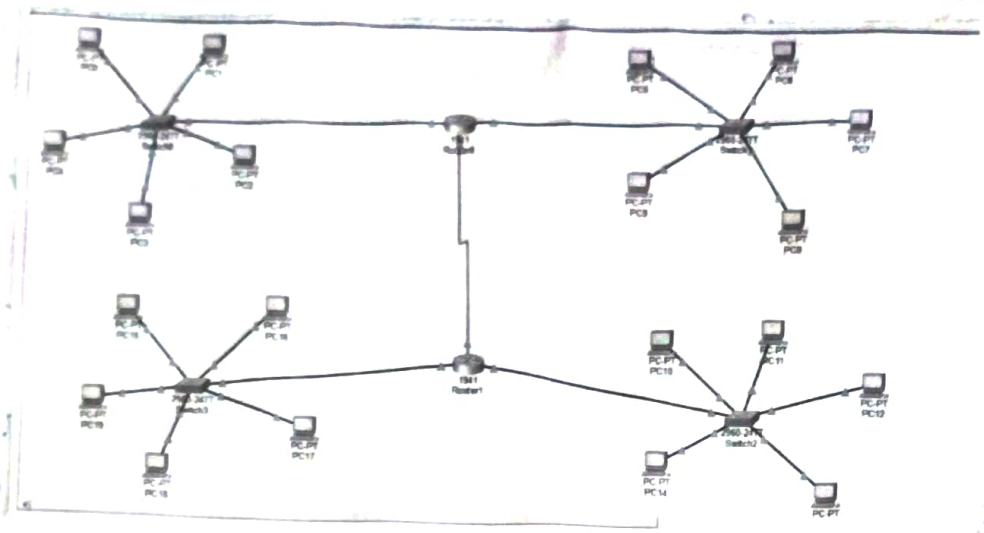
To implement classeless IP Subnetting and test network connectivity using cisco packet Tracer.

Procedure:

* Open Cisco packet Tracer and Create a new network topology.
* Add the required Routers, Switches, and PCs to the workspace.
* Connect the devices using appropriate cables.
* Dividing the network 192.168.1.0/24 into smaller subnets using a /27 Subnet mask.
* Assign Suitable IP address and Subnet masks to the router interfaces.
* configure the PCs with their respective IP address, Subnet Mask, and default gateway.
* Enable the router interfaces using the no shutdown command.
* Using ping command from the PCs to test connectivity.
* Verify that the packets are Successfully transmitted b/w the interfaces.

CLI command:

```
# enable
# configure terminal
# interface FastEthernet 0/0
# ip address 198.168.1.10 255.255.255.0
# no shutdown
# exit
```



Result:

Thus, classless IP subnetting was successfully implemented and tested using Cisco Packet Tracer.